The Binomial Theorem

Binomial coefficients

1 Evaluate each binomial coefficient:

a $\binom{6}{2}$ **b** $\binom{8}{0}$ **c** $\binom{5}{5}$ **d** $\binom{10}{3}$

2 Write down row 4 of Pascal's triangle (the row beginning $1, 4, \ldots$) and explain how it relates to the expansion of $(a+b)^4$.

Expansions and specific terms

3 Expand $(x+2)^4$ fully.

4 Find the coefficient of x^3 in the expansion of $(2x-1)^5$.

5 Find the term containing y^4 in the expansion of $(x+y)^7$.

6 Use the binomial theorem to expand $(1+0.02)^5$ to the x^2-style second-order term, and hence estimate 1.02^5 to four decimal places.